Biological Methods for Engineers (227-0949-00L) - Mock Exam Theory, Autumn Semester 2025

Combined question sheet and scoring rubric generated from JSON source. Criterion headings match rubric_manifest.json.

q1 - Topic: Lecture 01 — Introduction, Cell Culture, Biosafety Question 1:... - 1 point:

Question (q1): Topic: Lecture 01 — Introduction, Cell Culture, Biosafety

Question 1: What is the primary goal of a laboratory notebook, and why must even trivial observations be recorded?
Reference answer: Purpose of a Lab Notebook A laboratory notebook ensures experimental reproducibility by documenting every step, condition, unexpected event, and observation. Biology is not an exact science, and even subtle deviations can influence results. A properly kept notebook enables any other scientist to repeat the experiment without additional explanation.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q2 - Topic: Lecture 01 — Introduction, Cell Culture, Biosafety Question 2:... - 1 point:

Question (q2): Topic: Lecture 01 — Introduction, Cell Culture, Biosafety

Question 2: What distinguishes primary cells from continuous cell lines, and what biological mechanism explains why primary cells enter senescence?
Reference answer: Primary vs. Continuous Cell Lines Primary cells are derived directly from tissue and undergo a limited number of divisions before entering senescence. Continuous cell lines are immortalized due to genetic mutations that bypass senescence, permitting indefinite proliferation. Because of this, continuous lines are easier to culture but are more genetically altered.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q3 - Topic: Lecture 01 — Introduction, Cell Culture, Biosafety Question 3:... - 1 point:

Question (q3): Topic: Lecture 01 — Introduction, Cell Culture, Biosafety

Question 3: Why are CHO cells widely used in biotechnology, and which medium are they usually grown in?
Reference answer: Reasons CHO Cells Are Widely Used CHO cells are robust, easy to culture, and capable of high-yield expression of recombinant proteins. As mammalian cells, they support proper folding and many post-translational modifications. For biotechnology, their productivity makes them the functional mammalian counterpart to E. coli.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q4 - Topic: Lecture 01 — Introduction, Cell Culture, Biosafety Question 4:... - 1 point:

Question (q4): Topic: Lecture 01 — Introduction, Cell Culture, Biosafety

Question 4: What safety features define a Biosafety Cabinet Class II, and why must materials be sprayed with 70% ethanol before entering it?
Reference answer: Biosafety Cabinet Class II Principles Class II cabinets generate vertical laminar flow filtered through HEPA membranes, protecting the user, the environment, and the cultured sample. Materials entering the cabinet are sprayed with 70% ethanol to reduce microbial contamination.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q5 - Topic: Lecture 01 — Introduction, Cell Culture, Biosafety Question 5:... - 1 point:

Question (q5): Topic: Lecture 01 — Introduction, Cell Culture, Biosafety

Question 5: What is the difference between molarity and percentage solutions (w/v or v/v)?
Reference answer: Molarity vs. Percent Solutions Molarity (M) is defined as moles of solute per liter of solution. Percent solutions express ratios either as mass per volume (w/v: grams per 100 mL) or volume per volume (v/v: mL per 100 mL). They do not rely on molecular weight.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q6 - Topic: Lecture 02 — Protein Isolation & Immunoprecipitation Question... - 1 point:

Question (q6): Topic: Lecture 02 — Protein Isolation & Immunoprecipitation

Question 6: What is the functional difference between ionic and nonionic detergents when solubilizing proteins?
Reference answer: Ionic vs. Nonionic Detergents Ionic detergents such as SDS carry charged headgroups and disrupt both protein–protein and protein–lipid interactions, fully denaturing proteins. Nonionic detergents such as NP-40 solubilize membranes while largely preserving native protein complexes due to their uncharged hydrophilic groups.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q7 - Topic: Lecture 02 — Protein Isolation & Immunoprecipitation Question... - 1 point:

Question (q7): Topic: Lecture 02 — Protein Isolation & Immunoprecipitation

Question 7: In a typical whole-cell lysis buffer, why is EDTA included?
Reference answer: Function of EDTA in Lysis Buffers EDTA chelates divalent metal ions (e.g., $Mg2+$, $Ca2+$), inhibiting metal-dependent proteases and nucleases that would otherwise degrade proteins and nucleic acids during extraction.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q8 - Topic: Lecture 02 — Protein Isolation & Immunoprecipitation Question... - 1 point:

Question (q8): Topic: Lecture 02 — Protein Isolation & Immunoprecipitation

Question 8: Conceptually, how does the Bradford assay detect proteins?
Reference answer: Bradford Assay Detection Mechanism Coomassie Brilliant Blue G-250 shifts to its anionic (blue) form upon binding to proteins, resulting in increased absorbance at 595 nm. This absorbance is proportional to protein concentration when compared with a standard curve.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q9 - Topic: Lecture 02 — Protein Isolation & Immunoprecipitation Question... - 1 point:

Question (q9): Topic: Lecture 02 — Protein Isolation & Immunoprecipitation

Question 9: What role does SDS play in SDS-PAGE, and how does this enable size-based separation?
Reference answer: Role of SDS in SDS-PAGE SDS binds uniformly along the protein backbone, imparting a consistent negative charge proportional to protein length while denaturing tertiary structure. Proteins then migrate solely based on size through the polyacrylamide matrix.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q10 - Topic: Lecture 02 — Protein Isolation & Immunoprecipitation Question... - 1 point:

Question (q10): Topic: Lecture 02 — Protein Isolation & Immunoprecipitation

Question 10: Why is an antibody bound to streptavidin-coated magnetic beads used during immunoprecipitation?
Reference answer: Principle of Streptavidin-Coated Magnetic Beads Biotinylated antibodies bind streptavidin-coated magnetic beads with extremely high affinity, allowing antigen– antibody complexes to be isolated magnetically from the lysate during immunoprecipitation.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q11 - Topic: Lecture 03 — Genome Engineering Question 11: What are the two... - 1 point:

Question (q11): Topic: Lecture 03 — Genome Engineering

Question 11: What are the two major natural DNA repair pathways that respond to a doublestrand break, and which one is typically used for gene knockout?
Reference answer: NHEJ vs. Homologous Recombination Nonhomologous end joining (NHEJ) rapidly repairs double-strand breaks without a template and is error-prone, often generating small deletions or insertions. Homologous recombination (HR) uses a donor template to repair breaks precisely, enabling controlled knock-ins.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q12 - Topic: Lecture 03 — Genome Engineering Question 12: Why was the devel... - 1 point:

Question (q12): Topic: Lecture 03 — Genome Engineering

Question 12: Why was the development of zinc-finger nucleases considered an early breakthrough for targeted genome editing?

Reference answer: Significance of Zinc-Finger Nucleases Zinc-finger nucleases combine engineered DNA-binding domains with the FokI nuclease, enabling the first programmable, sequencespecific genome editing in mammalian cells. They represented a major step toward modern CRISPR systems.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q13 - Topic: Lecture 03 — Genome Engineering Question 13: What two componen... - 1 point:

Question (q13): Topic: Lecture 03 — Genome Engineering

Question 13: What two components are required for CRISPR–Cas9 genome editing in mammalian cells?
Reference answer: Required Components for CRISPR–Cas9 Editing CRISPR editing in mammalian cells requires Cas9 nuclease and a guide RNA (gRNA). The gRNA directs Cas9 to the target sequence via base complementarity.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q14 - Topic: Lecture 03 — Genome Engineering Question 14: What is a PAM seq... - 1 point:

Question (q14): Topic: Lecture 03 — Genome Engineering

Question 14: What is a PAM sequence, and why is it essential for Cas9 activity?
Reference answer: PAM Sequence and Its Role Cas9 requires a short PAM sequence (for SpCas9, the motif NGG) adjacent to the target DNA. Cas9 binds the PAM and introduces a double-strand break approximately three nucleotides upstream of it.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q15 - Topic: Lecture 03 — Genome Engineering Question 15: Why does Cas9-ind... - 1 point:

Question (q15): Topic: Lecture 03 — Genome Engineering

Question 15: Why does Cas9-induced double-strand break often result in small deletions (indels)?
Reference answer: Why Cas9 Creates Small Indels Following Cas9 cleavage, NHEJ repairs the double-strand break without a template, often removing or inserting a few nucleotides due to exonuclease trimming of DNA ends. These indels frequently disrupt coding sequences, generating knockouts.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q16 - Topic: Lecture 04 — Induced Neurons Question 16: What distinguishes a... - 1 point:

Question (q16): Topic: Lecture 04 — Induced Neurons

Question 16: What distinguishes a neuron from most other cell types in terms of structure and function?
Reference answer: Key Properties of Neurons Neurons possess structurally distinct dendrites and a long axon, enabling directional signal transmission. Their ability to form synaptic networks and generate action potentials distinguishes them from most other cell types.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q17 - Topic: Lecture 04 — Induced Neurons Question 17: What is the purpose... - 1 point:

Question (q17): Topic: Lecture 04 — Induced Neurons

Question 17: What is the purpose of seeding neurons into PDMS microstructures?
Reference answer: Purpose of PDMS Microstructures for iNeurons PDMS microstructures physically guide axonal outgrowth, enforcing connectivity patterns (e.g., directional or circular). They are used to build defined neural circuits for bottom-up neuroscience experiments.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q18 - Topic: Lecture 04 — Induced Neurons Question 18: What key discovery b... - 1 point:

Question (q18): Topic: Lecture 04 — Induced Neurons

Question 18: What key discovery by Shinya Yamanaka enabled the creation of induced pluripotent stem cells (iPSCs)?
Reference answer: Yamanaka Reprogramming Factors Shinya Yamanaka demonstrated that expression of four transcription factors (Oct4, Sox2, Klf4, c-Myc) reprograms differentiated cells into induced pluripotent stem cells (iPSCs), restoring pluripotency.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q19 - Topic: Lecture 04 — Induced Neurons Question 19: What is the role of... - 1 point:

Question (q19): Topic: Lecture 04 — Induced Neurons

Question 19: What is the role of Matrigel in iNeuron culture?
Reference answer: Role of Matrigel Matrigel is a complex extracellular matrix extract providing laminin, collagen IV, and growth factors. It promotes adhesion and survival of iNeurons, which do not readily attach to untreated surfaces.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q20 - Topic: Lecture 04 — Induced Neurons Question 20: Conceptually, how do... - 1 point:

Question (q20): Topic: Lecture 04 — Induced Neurons

Question 20: Conceptually, how does Calcein AM differentiate live cells from dead cells?
Reference answer: Mechanism of Live/Dead Staining Calcein-AM diffuses into cells and is converted by intracellular esterases into fluorescent calcein, marking live cells in green. Ethidium homodimer-1 cannot cross intact membranes and enters only dead cells, binding DNA and fluorescing red.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q21 - Topic: Lecture 05 — qRT-PCR Question 21: Why is quantifying mRNA leve... - 1 point:

Question (q21): Topic: Lecture 05 — qRT-PCR

Question 21: Why is quantifying mRNA levels a useful proxy for gene expression?
Reference answer: Why mRNA Quantification Reflects Gene Expression Because transcription is a major regulatory step, mRNA abundance frequently correlates with gene activity. Measuring mRNA levels provides a fast, efficient proxy for gene expression changes.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q22 - Topic: Lecture 05 — qRT-PCR Question 22: Why is Nanodrop absorbance a... - 1 point:

Question (q22): Topic: Lecture 05 — qRT-PCR

Question 22: Why is Nanodrop absorbance at 260 nm used to quantify nucleic acids?
Reference answer: Why Nucleic Acids Absorb at 260nm The aromatic nucleotide bases absorb UV light maximally at 260 nm, enabling quantification of RNA and DNA using spectrophotometry.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q23 - Topic: Lecture 05 — qRT-PCR Question 23: What is the purpose of using... - 1 point:

Question (q23): Topic: Lecture 05 — qRT-PCR

Question 23: What is the purpose of using a poly-T primer during reverse transcription?
Reference answer: Purpose of Poly-T Primers Poly-T primers anneal to the poly-A tails of mature mRNA, ensuring that reverse transcription selectively targets mRNA rather than rRNA or tRNA.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q24 - Topic: Lecture 05 — qRT-PCR Question 24: What defines the cycle thres... - 1 point:

Question (q24): Topic: Lecture 05 — qRT-PCR

Question 24: What defines the cycle threshold (CT) in qPCR?
Reference answer: Definition of Cycle Threshold (CT) The CT is the PCR cycle at which fluorescence surpasses a predefined threshold during exponential amplification. Lower CT values indicate higher initial template abundance.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q25 - Topic: Lecture 05 — qRT-PCR Question 25: Why must expression levels o... - 1 point:

Question (q25): Topic: Lecture 05 — qRT-PCR

Question 25: Why must expression levels of a target gene be normalized to a housekeeping gene?
Reference answer: Reason for Housekeeping Gene Normalization Housekeeping genes provide a stable internal reference to correct for variability in RNA input, cDNA synthesis, or pipetting. This ensures accurate relative quantification of gene expression.

Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q26 - Topic: Lecture 06 — Transfection & RNAi Question 26: Conceptually, wh... - 1 point:

Question (q26): Topic: Lecture 06 — Transfection & RNAi

Question 26: Conceptually, what is the difference between transient and stable transfection?
Reference answer: Transient vs. Stable Transfection Transient transfection introduces plasmid DNA that remains episomal and expresses transiently. Stable transfection requires genomic integration, producing long-term, heritable expression.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q27 - Topic: Lecture 06 — Transfection & RNAi Question 27: Why do cationic... - 1 point:

Question (q27): Topic: Lecture 06 — Transfection & RNAi

Question 27: Why do cationic polymers (like PEI) enable chemical transfection?
Reference answer: How PEI Enables Transfection PEI forms positively charged complexes with DNA that bind to negatively charged cell membranes and enter via endocytosis. Its buffering capacity promotes endosomal escape (“proton sponge effect”), releasing DNA into the cytosol.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q28 - Topic: Lecture 06 — Transfection & RNAi Question 28: Why is long doub... - 1 point:

Question (q28): Topic: Lecture 06 — Transfection & RNAi

Question 28: Why is long double-stranded RNA toxic to mammalian cells, and how does this affect RNA interference experiments?
Reference answer: Why Long dsRNA Is Toxic in Mammals Long dsRNA activates the interferon and PKR pathways, resulting in global translational shutdown and apoptosis. Therefore, mammalian RNAi uses short siRNAs to avoid innate immune activation.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q29 - Topic: Lecture 06 — Transfection & RNAi Question 29: What is the func... - 1 point:

Question (q29): Topic: Lecture 06 — Transfection & RNAi

Question 29: What is the functional role of the RISC complex in RNAi?
Reference answer: Functional Role of RISC RISC loads the guide strand of siRNA via Argonaute, identifies complementary mRNA, and induces cleavage (with perfect pairing) or translational repression (with partial pairing).
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q30 - Topic: Lecture 06 — Transfection & RNAi Question 30: Why are genome-w... - 1 point:

Question (q30): Topic: Lecture 06 — Transfection & RNAi

Question 30: Why are genome-wide RNAi screens often performed in multi-well plates?
Reference answer: Why Genome-Wide RNAi Screens Use Multiwell Plates Multiwell formats enable high-throughput parallel testing of thousands of siRNAs, minimizing crosscontamination and allowing automated quantitative readouts.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q31 - Topic: Lecture 07 — DNA Electrophoresis & XBP1 Splicing Question 31:... - 1 point:

Question (q31): Topic: Lecture 07 — DNA Electrophoresis & XBP1 Splicing

Question 31: Why does DNA migrate in an electric field during gel electrophoresis?
Reference answer: Why DNA Migrates in an Electric Field DNA has a uniformly negative phosphate backbone and migrates toward the positive electrode during electrophoresis.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q32 - Topic: Lecture 07 — DNA Electrophoresis & XBP1 Splicing Question 32:... - 1 point:

Question (q32): Topic: Lecture 07 — DNA Electrophoresis & XBP1 Splicing

Question 32: What determines the choice between agarose and polyacrylamide gels for DNA separation?
Reference answer: Agarose vs. Polyacrylamide Separation Agarose gels suit large DNA fragments; polyacrylamide gels resolve small fragments with high precision. Gel percentage determines pore size and resolution.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q33 - Topic: Lecture 07 — DNA Electrophoresis & XBP1 Splicing Question 33:... - 1 point:

Question (q33): Topic: Lecture 07 — DNA Electrophoresis & XBP1 Splicing

Question 33: What is the biochemical effect of tunicamycin, and how does it induce ER stress?
Reference answer: Biochemical Effect of Tunicamycin Tunicamycin inhibits N-linked glycosylation by blocking GlcNAc phosphotransferase, causing misfolded protein accumulation and ER stress.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q34 - Topic: Lecture 07 — DNA Electrophoresis & XBP1 Splicing Question 34:... - 1 point:

Question (q34): Topic: Lecture 07 — DNA Electrophoresis & XBP1 Splicing

Question 34: What is unconventional cytoplasmic splicing of XBP-1, and what functional change does it create?
Reference answer: Unconventional Splicing of XBP1 The endonuclease IRE1 cleaves XBP1 mRNA in the cytosol, removing a 26-nucleotide intron. The resulting frameshift produces XBP1s, a potent transcription factor in the UPR.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q35 - Topic: Lecture 07 — DNA Electrophoresis & XBP1 Splicing Question 35:... - 1 point:

Question (q35): Topic: Lecture 07 — DNA Electrophoresis & XBP1 Splicing

Question 35: Why does the XBP-1 unspliced and spliced PCR product differ in size by 26 nucleotides?
Reference answer: Why XBP1 PCR Products Differ by 26 Nucleotides The XBP1 mRNA contains a 26-nucleotide unconventional intron, which is removed by the endonuclease IRE1 during ER stress. This cytoplasmic splicing event generates a shorter PCR product for the spliced isoform (XBP1s) and a longer product for the unspliced form (XBP1u), resulting in a 26-nucleotide difference in electrophoretic mobility.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.